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Summary

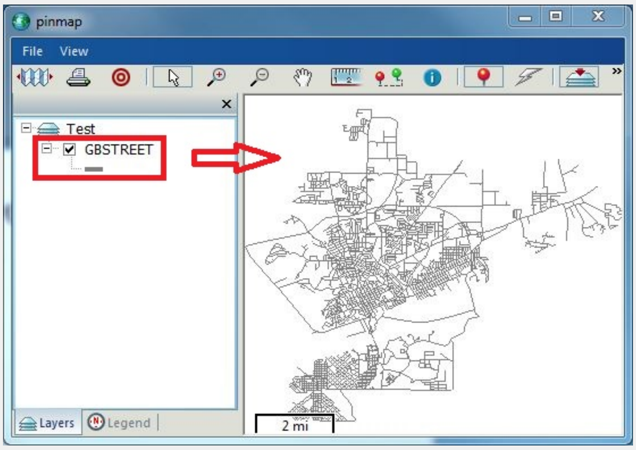
Covers how to distribute updated map files (shapefiles) via media, network shares, and automation through Mobile / Message Center

Details

What are Shapefiles?

Shapefiles are one of several GIS industry standard file structures used to store visual geographic (map) data. Several files, each with the same name but unique extensions, are used to represent one layer on a map. Therefore, while there are a few occasions where it is appropriate to refer to a 'shapefile' (singular), it is usually more accurate to refer to 'shapefiles' (plural) as a set, such as "Can you get me the shapefiles for that layer?" Hence, when updating a layer, the entire updated set of files must be provided. Furthermore, in order to integrate properly with your Spillman data, all layers must be in a GCS WGS84 or GCS NAD83 coordinate system (projection).

Example:
This map has one layer, named GBSTREET



The data needed to display the GBSTREET layer
is stored in the following set of files

GBSTREET.dbf
GBSTREET.prj
GBSTREET.sbn
GBSTREET.sbx
GBSTREET.shp
GBSTREET.shp.xml
GBSTREET.shx

Distribution Option 1: Media

The most basic way to distribute an updated set of shapefiles is to use media such as a thumb drive, SD card, CD, or DVD. Save the updated files from ArcMap onto media, take the media to each Spillman PC, and copy the files into the Spillman map path on that machine. (How to find the map path is illustrated below)

Distribution Option 2: Copy from Network Share

A slightly more sophisticated way to get updated files onto each client PC is to copy the new files from a network share location. This is still manual, PC by PC, and can be done from anywhere on the network with sufficient permissions. This could be automated by server admins through the use of scripts / batch files and scheduled events, or pushed out through Active Directory in Windows Server environments.

Distribution Option 3: Point Map Path to Network Share (NOT RECOMMENDED)

Shapefiles may be placed in a shared network folder and referenced in the Mobile or Spillman map path (below) with a UNC such as \\AgencyFileServer\SpillmanMaps. With this method, since all clients would be pointed to the same files on the network, updates may be accomplished through simply replacing those files in that one network location. However, getting them replaced is not all that simple. When a map is open on any client, shapefiles are in use and are therefore locked by the operating system. You cannot overwrite them while they are in use. Thus, in order to get those files replaced, you would have to be able to either get everyone to exit their maps before updating the files, or you'd have to forcefully close those connections through server administration tools. Also, pointing mobile clients to an internal network path would provide at least a very slow, if not an impossible configuration in your environment. Even using this method for just internal PCs running full Spillman, the map performance will be slower than if the files were located locally on each machine.

Distribution Option 4: Pull through Mobile / Message Center from Spillman Server (RECOMMENDED)

Mobile / Message Center is designed to always look on the Spillman Server for updated map files at each login. Those files must simply be placed in the appropriate location on the Spillman Server (see below) for this to happen; there is no further configuration. Any new filenames or files with newer date/time stamps will be pulled over the network and saved into the client map path (see below). If it is Message Center opening (such as part of opening full Spillman), it will not ask - it will just do this as part of logging in (it assumes a hard-line connection). If it is Mobile that is opening and new files are found on the Spillman Server, Mobile will prompt the user to Update Now, Customize, or Ask Me Later. Update Now will pull the new files and save them to the client map path. Customize allows the user to scrutinize the list of files being offered, and check or uncheck those wanted.

Map files may be distributed to all users, per state, per agency, or per user by placing shapefiles at the following paths on the Spillman Server:

Distribution Level	Example Unix / Linux Path	Example Windows Path
All	$MOBILEDIR/uclient/all/maps	%MOBILEDIR%/uclient/all/maps
State	$MOBILEDIR/uclient/ut/maps	%MOBILEDIR%/uclient/ut/maps
Agency	$MOBILEDIR/uclient/spd/maps	%MOBILEDIR%/uclient/spd/maps
User	$MOBILEDIR/uclient/train1/maps	%MOBILEDIR%/uclient/train1/maps

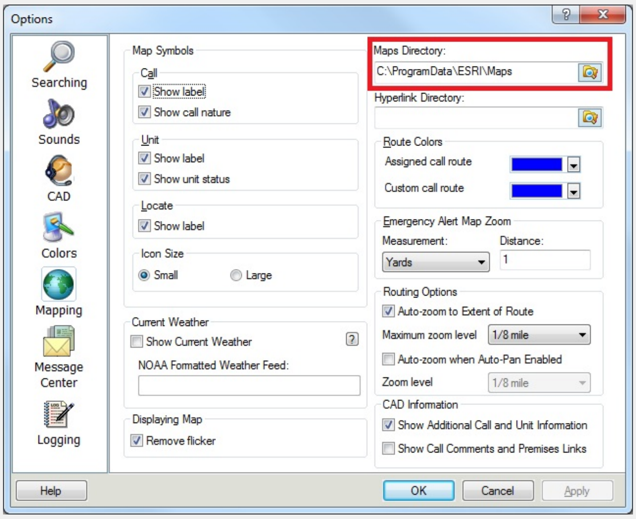
In order to determine the value of the environmental variable MOBILEDIR on your system, type echo $MOBILEDIR (Unix / Linux) or echo %MOBILEDIR% (Windows) in a console / command window.

If any of the folders in the path do not exist on your server, create them and grant all users **full** permissions to them.

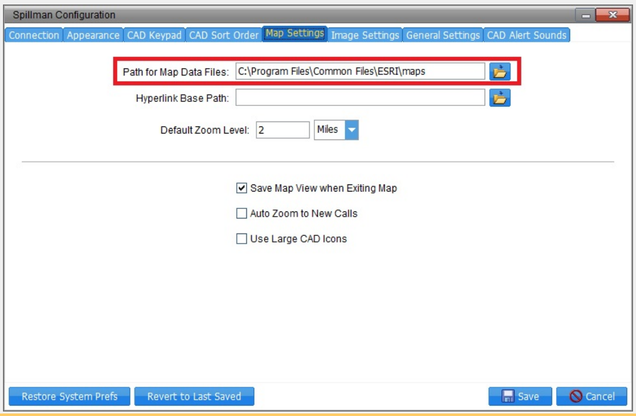
Tips

- In most cases, you do **NOT** want to include the .gcd file (like GBSTREET.gcd) in your distribution. It is generated dynamically, and putting an old one in place may render the Locate Tool inoperable until you remove the .gcd file from the local PC and let Spillman generate a new one.
- All distribution level folder names must be in lower case (i.e., "all" and "ut" for state "UT" and “spd” for agency “SPD”).
- The agency code keys off the agency listed in cdunit, not syunit or apnames.
- If the map path for both Spillman and Mobile / Message Center match, then the maps are being updated for all three maps types: CAD Map, Pin Map, and Mobile Map. Mobile is pulling the updated files from the server and saving them to one local folder at which all three maps are pointing to get their data.
- The main consideration with Option 4 is connection speed in mobile units. That is why the mobile user has the option to decline the update (Ask Me Later); they can choose to receive it at a later time when they've established a faster connection.

Finding/Setting map path in Mobile: File –> Options –> Mapping



Finding/Setting map path in Spillman: File --> Configuration --> Map Settings



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